

1899–1975

PERCY JULIAN

African American Percy Julian overcame racial barriers to be one of the greatest research chemists of the 20th century.

As a research fellow at DePauw University, Percy Julian was the first to chemically produce the anti-glaucoma drug physostigmine, making it widely available. Racial prejudice forced him out of DePauw, and he moved to a soybean oil company. Here, he pioneered the large-scale synthesis of human hormones from plants and became a world-class research scientist.

Mass drug production

In 1940, Julian devised a cost-effective way to extract the steroid stigmasterol from soybean oil and convert it into the hormone progesterone on an industrial scale. His work led to the synthesis of many other hormones from natural sources and helped to secure their widespread use.

MILESTONES

CHEMISTRY PhD

Graduates from DePauw as valedictorian in 1920; gains PhD in chemistry in 1931 from University of Vienna.

DRUG SYNTHESIS

Achieves a total synthesis of physostigmine in 1935; publishes results in a landmark research paper.

HUMAN HORMONES

Leads key research into the synthesis of human hormones from plant products in the 1940s.

PLANT STEROIDS

Sets up own company, Julian Laboratories, in 1953, continuing in the synthesis of plant steroids.

NATIONAL ACCLAIM

Becomes the first African American chemist elected to the National Academy of Sciences in 1973.

“We **dared** to do something rather **bold**.”

Percy Julian, 1965

The discovery of stigmasterol as a by-product of soybean oil resulted from an accident. Water leaked into a tank of oil, creating a white solid: Julian identified it as the plant-derived steroid.

